

Question 1

Which IEEE 802.1X role is responsible for making the decision to allow or deny a supplicant access to the LAN and placing it into the appropriate VLAN once authentication is successful?

- A. Access Point
- B. Authentication Server
- C. Authenticator
- D. Supplicant

Answer:

B. Authentication Server

Explanation:

In **IEEE 802.1X**, the Authentication Server (typically a RADIUS server) is responsible for validating credentials and making the final access decision. It determines whether the supplicant is allowed or denied access and can also assign network policies such as VLAN placement after successful authentication.

The Authenticator (switch or access point) only forwards authentication requests and enforces the decision, while the Supplicant is simply the client requesting access.

Explanation (Simplified):

In IEEE 802.1X, there are three roles:

- Supplicant → The client requesting access
- Authenticator → The switch/AP that passes credentials
- Authentication Server → The decision-maker

👉 The Authentication Server (usually a RADIUS server):

- Validates credentials
- Decides allow or deny access
- Assigns policies like VLAN placement

Quick Memory Trick:

- Supplicant = Requests
- Authenticator = Middleman
- Authentication Server = Brain (decision) 🧠

Key Point:

Authentication Server = Decision maker (Allow/Deny + VLAN assignment)

Question 2

You need to configure a router to prioritize VoIP traffic over regular data traffic. Which of the following configurations is most appropriate to achieve this using QoS on a Cisco router?

- A. class-map match-any VOIP; match protocol RTP
- B. ip route 0.0.0.0 0.0.0.0 192.168.1.1
- C. access-list 101 permit ip any any
- D. policy-map VOIP-PRIORITY; class VOIP; priority 1000

Answer:

D. policy-map VOIP-PRIORITY; class VOIP; priority 1000 ✅

Explanation:

In Cisco QoS, a **policy-map** is used to define how traffic should be treated. The command **priority** gives **strict priority (low latency)** to traffic, which is essential for VoIP. This ensures voice packets are sent first, reducing delay, jitter, and packet loss.

- **class-map** → Identifies traffic (e.g., RTP for VoIP)
- **policy-map** → Applies actions (like priority, bandwidth)
- **priority command** → Guarantees low-latency queue for VoIP

Other options:

- **A** → Only identifies VoIP traffic, does not prioritize it
- **B** → Default route, unrelated to QoS

- **C** → Permits all traffic, no prioritization

Explanation (Simplified):

In QoS, there are 3 main steps:

- Classify → Identify traffic (VoIP, video, data)
- Mark → Tag traffic (DSCP, CoS)
- Prioritize → Give importance (QoS policy)

👉 The **policy-map with priority command**:

- Gives VoIP highest priority
- Reduces delay and jitter
- Ensures smooth calls

🧠 Quick Memory Trick:

- class-map = Identify traffic
- policy-map = Apply rules
- priority = **Real-time traffic first (VoIP)** 🎤

Key Point:

VoIP needs **low latency** → use **priority** in policy-map 🚀

Question 3

Which of the following methods would reduce the broadcast domain size in a large Layer 2 switched network?

- A. Port Security
- B. Spanning Tree Protocol
- C. Subnetting
- D. VLANs

Answer:

- D. VLANs ✅

Explanation:

VLANs (Virtual Local Area Networks) logically divide a Layer 2 network into multiple smaller broadcast domains. Each VLAN acts as its own separate network, so broadcasts are contained within that VLAN instead of flooding the entire switch.

- **Port Security** → Controls device access, not broadcast size
- **Spanning Tree Protocol (STP)** → Prevents loops, not broadcast reduction
- **Subnetting** → Works at Layer 3, not directly at Layer 2 broadcast control
- **VLANs** → Directly reduce broadcast domains at Layer 2 

Explanation (Simplified):

In a Layer 2 network:

- All devices in the same VLAN = same broadcast domain
- More VLANs = smaller broadcast domains

 VLANs:

- Split one big network into smaller ones
- Reduce unnecessary broadcast traffic
- Improve performance and security

Quick Memory Trick:

- VLAN = **Virtual separation of networks**
- More VLANs = **Less broadcast traffic** 

Key Point:

VLANs = Reduce broadcast domain size in Layer 2 networks 

Question 4

Which of the following options correctly describes the function of a router on a network?

A. Routers determine the best path to forward packets based on the destination IP address.

- B. Routers segment a network into different collision domains.
- C. Routers operate at the Data Link layer and mainly handle MAC addresses.
- D. Routers convert IP addresses to MAC addresses using ARP.

Answer:

A. Routers determine the best path to forward packets based on the destination IP address. 

Explanation:

Routers operate at **Layer 3 (Network Layer)** of the OSI model. Their primary function is to **analyze destination IP addresses and determine the best path** to forward packets between different networks. They use routing tables and protocols (like OSPF, BGP) to make these decisions.

Other options:

- **B** → Collision domains are handled by switches/bridges
- **C** → Data Link layer & MAC addresses belong to switches (Layer 2)
- **D** → ARP is used to map IP to MAC, but this is not the main function of a router

Explanation (Simplified):

- Router = Connects different networks
- Uses IP address to decide where packets go

 Router does:

- Reads destination IP
- Checks routing table
- Forwards packet to best path

Quick Memory Trick:

- Router = **Path chooser (IP-based)** 
- Switch = MAC-based (Layer 2)
- Router = IP-based (Layer 3)

Key Point:

Router = Finds best path using destination IP address 🚀

Question 5

In a network environment where both 802.1X and MAB (MAC Authentication Bypass) are implemented, how can a network administrator ensure that the devices that do not support 802.1X are still granted network access?

- A. By using port-based VLAN assignment to segregate devices and apply appropriate access policies
- B. By configuring a fallback mechanism to allow non-802.1X devices in periodically
- C. By configuring the switch ports to use both 802.1X and MAB, with MAB acting as a fallback method
- D. By disabling 802.1X on ports where non-802.1X devices are connected

Answer:

C. By configuring the switch ports to use both 802.1X and MAB, with MAB acting as a fallback method ✅

Explanation:

In **IEEE 802.1X**, devices authenticate using credentials. However, some devices (like printers, IP phones, IoT devices) do not support 802.1X.

To handle this, **MAB (MAC Authentication Bypass)** is used as a fallback. If a device fails 802.1X authentication or does not respond, the switch uses the device's MAC address to authenticate it via a RADIUS server.

Other options:

- **A** → VLAN assignment helps segmentation, not authentication fallback
- **B** → No such standard “periodic fallback” mechanism
- **D** → Disabling 802.1X reduces security (not recommended)

Explanation (Simplified):

In secure networks:

- First try → **802.1X authentication (username/cert)**
- If not supported → **Fallback to MAB (MAC-based)**

👉 MAB allows:

- Printers, IP phones, IoT devices to connect
- Authentication using MAC address
- Continued security without disabling 802.1X

🧠 Quick Memory Trick:

- 802.1X = Smart devices (login-based)
- MAB = **Fallback for “dumb” devices (MAC-based)** 📩

Key Point:

Use **802.1X first + MAB fallback** to support all devices securely 🚀

Question 6

You have been asked to configure a new VLAN on a Cisco switch. What command would you use to assign an interface to VLAN 10 and set it to be an access port?

- A. switchport access vlan 10
- B. switchport mode access and switchport access vlan 10
- C. switchport mode trunk
- D. vlan database vlan 10

Answer:

- B. switchport mode access and switchport access vlan 10 ✅

Explanation:

To correctly assign a port to a VLAN on a Cisco switch, you must:

1. Set the port to **access mode**
2. Assign it to the desired VLAN

The correct configuration requires **both commands together**:

- `switchport mode access` → Sets the port as an access port
- `switchport access vlan 10` → Assigns the port to VLAN 10

Other options:

- **A** → Assigns VLAN but does not ensure access mode
- **C** → Configures trunk port (used between switches, not end devices)
- **D** → Old/deprecated method, not used in modern configs

Explanation (Simplified):

For end devices (PC, printer):

- Use **access port**
- Assign VLAN

👉 Required steps:

- Set mode = access
- Set VLAN = 10

🧠 Quick Memory Trick:

- Access port = **1 VLAN only (end device)**
- Trunk port = Multiple VLANs (switch-to-switch)

👉 Always:

mode access + access vlan = correct config ✅

Key Point:

Access port configuration =

switchport mode access + switchport access vlan X 🚀

Question 7

Which command would you use to configure a VLAN membership policy on a Cisco switch to dynamically assign VLANs based on MAC addresses?

- A. switchport trunk allowed vlan add
- B. mac-address-table static
- C. vlan-assignment policy
- D. vmps

Answer:

D. vmps 

Explanation:

VMPS (VLAN Membership Policy Server) is used on Cisco switches to dynamically assign VLANs to devices based on their MAC addresses. When a device connects, the switch queries the VMPS server, which returns the appropriate VLAN assignment.

Other options:

- **A** → Used to allow VLANs on trunk ports (not dynamic assignment)
- **B** → Statically maps MAC addresses, not VLAN assignment
- **C** → Not a valid Cisco command for VLAN assignment

Explanation (Simplified):

Dynamic VLAN assignment works like this:

- Device connects → switch reads MAC address
- Switch asks VMPS server
- VMPS assigns VLAN automatically

 Useful for:

- Large networks
- Automatic user/device placement
- Centralized VLAN control

Quick Memory Trick:

- VMPS = **VLAN by MAC address** 
- Static VLAN = Manual assignment
- VMPS = Automatic assignment

Key Point:

VMPS = Dynamic VLAN assignment using MAC addresses 

Question 8

When configuring an IPv6 address on an interface, which specific type of address is automatically generated and what is its primary purpose?

- A. Anycast address
- B. Global unicast address
- C. Unique local address
- D. Link-local address

Answer:

- D. Link-local address 

Explanation:

In IPv6, a **link-local address** is automatically generated on every interface, even if no manual configuration is performed. It typically starts with **FE80::/10** and is used for communication within the same local network segment (link).

This address is essential for basic network operations such as:

- Neighbor Discovery Protocol (NDP)
- Router discovery
- Local communication between devices

Other options:

- **A (Anycast)** → Used to reach the nearest device in a group

- **B (Global unicast)** → Manually or automatically assigned for internet communication
- **C (Unique local)** → Private IPv6 addressing (similar to private IPv4)

Explanation (Simplified):

Every IPv6 device automatically gets:

- **Link-local address (FE80::)**

👉 Purpose:

- Talk to devices on same network
- Used for routing and discovery
- Works even without internet or DHCP

🧠 Quick Memory Trick:

- Link-local = **Local communication only** 🏠
- Starts with **FE80::**
- Always auto-generated

Key Point:

Link-local address = Automatic + required for local IPv6 communication 🚀

Question 9

In which scenario would you most likely use the IEEE 802.1Q standard in a network setup?

- To optimize the routing information exchanged between routers using an advanced algorithm.
- To establish a secure VPN tunnel across an enterprise network.
- To enable multiple VLANs to communicate over a single trunk connection.
- To manage and allocate IP addresses dynamically across multiple subnets in a network.

Answer:

C. To enable multiple VLANs to communicate over a single trunk connection. 

Explanation:

IEEE 802.1Q is a VLAN tagging protocol used on trunk links between switches (or switch-to-router). It inserts a tag into Ethernet frames to identify which VLAN the traffic belongs to.

This allows **multiple VLANs to share a single physical link**, making network design more efficient and scalable.

Other options:

- **A** → Refers to routing protocols (like OSPF, BGP)
- **B** → Refers to VPN technologies (IPsec, SSL)
- **D** → Refers to DHCP (IP address allocation)

Explanation (Simplified):

Without 802.1Q:

- One cable = One VLAN

With 802.1Q (trunk):

- One cable = **Multiple VLANs**

 It works by:

- Adding VLAN tag to frames
- Identifying which VLAN traffic belongs to
- Allowing switches to separate traffic correctly

Quick Memory Trick:

- 802.1Q = “**Q = Queue of VLANs on one link**” 
- Trunk port = Multiple VLANs
- Access port = Single VLAN

Key Point:

802.1Q = VLAN tagging → multiple VLANs over one trunk link 🚀

Question 10

A network administrator has configured a static route on a router for IP connectivity between two LANs. A traceroute from one LAN to the other shows that the packets are being sent to the wrong next-hop router, which leads to routing loops. What is the most likely cause of this issue?

- A. The administrative distance for the static route is too high.
- B. The IP address of the next-hop router is incorrectly configured.
- C. The subnet mask for the static route is incorrect.
- D. There is a duplicate static route with a different next-hop address.

Answer:

B. The IP address of the next-hop router is incorrectly configured. ✅

Explanation:

In static routing, the **next-hop IP address** tells the router where to forward packets. If this address is incorrect, traffic will be sent to the wrong router, which can cause **routing loops** and failed delivery.

Traceroute showing packets going to the wrong device strongly indicates a **misconfigured next-hop IP**.

Other options:

- **A** → Administrative distance affects route preference, not incorrect forwarding in a single static route
- **C** → Wrong subnet mask may cause reachability issues, but not specifically wrong next-hop routing loops
- **D** → Duplicate routes could cause confusion, but the question points directly to wrong next-hop behavior

Explanation (Simplified):

Static route works like:

- Destination → Next-hop → Forward packet

👉 If next-hop is wrong:

- Packet goes wrong direction
- May loop between routers
- Destination never reached

🧠 Quick Memory Trick:

- Static route = “Go to **THIS** router” 🗺️
- Wrong next-hop = **Wrong direction** = Loop 🔄

Key Point:

Incorrect next-hop IP = Packets forwarded incorrectly → Routing loops 🚀

Question 11

In a network utilizing both IPv4 and IPv6 addressing, which type of IPv6 address allows an IPv6-only network to communicate with an IPv4-only network, ensuring address translation and compatibility within the transition mechanism?

- A. Global unicast address
- B. IPv4-mapped IPv6 address
- C. Multicast address
- D. Link-local address

Answer:

- B. IPv4-mapped IPv6 address ✅

Explanation:

An **IPv4-mapped IPv6 address** is used to represent an IPv4 address within an IPv6 format (e.g., `::ffff:192.168.1.1`). This allows IPv6 systems to communicate with IPv4 systems during transition phases by enabling compatibility and translation between the two protocols.

These addresses are commonly used in dual-stack environments and transition mechanisms to support interoperability between IPv4 and IPv6 networks.

Other options:

- **A** → Used for global IPv6 communication, not IPv4 compatibility
- **C** → Used for one-to-many communication in IPv6
- **D** → Used only for local network communication

Explanation (Simplified):

During IPv4 → IPv6 transition:

- IPv4-mapped IPv6 = **IPv4 inside IPv6 format**

👉 Example:

- IPv4: 192.168.1.1
- IPv6: `::ffff:192.168.1.1`

👉 Purpose:

- Allows IPv6 devices to talk to IPv4 devices
- Helps in compatibility and transition

🧠 Quick Memory Trick:

- IPv4-mapped = “**IPv4 hidden inside IPv6**” 🔄
- Used for transition between IPv4 ↔ IPv6

Key Point:

IPv4-mapped IPv6 = Bridge between IPv6 and IPv4 networks 🚀

Question 12

Which of the following security protocols is used to provide encryption and confidential transmission over IP and is an integral part of IPsec?

- A. SSL (Secure Sockets Layer)
- B. AH (Authentication Header)
- C. ESP (Encapsulating Security Payload)
- D. IKE (Internet Key Exchange)

Answer:

- C. ESP (Encapsulating Security Payload) 

Explanation:

IPsec uses two main protocols:

- **AH (Authentication Header)** → Provides integrity and authentication (NO encryption)
- **ESP (Encapsulating Security Payload)** → Provides **encryption, confidentiality, integrity, and authentication**

👉 Therefore, **ESP** is responsible for encrypting data and ensuring secure transmission over IP networks.

Other options:

- **A (SSL)** → Used for web security (HTTPS), not part of IPsec
- **B (AH)** → Does NOT provide encryption
- **D (IKE)** → Used for key exchange, not data encryption

Explanation (Simplified):

In IPsec:

- AH = Integrity only  (no encryption)
- ESP = **Encryption + Security** 
- IKE = Key setup

👉 If question says **encryption/confidentiality** → always **ESP**

🧠 Quick Memory Trick:

- ESP = **Encrypts Secure Packets** 🔒
- AH = Authentication only
- IKE = Key exchange

Key Point:

ESP = Encryption + confidentiality in IPsec 🚀

Question 13

Which of the following statements accurately describes the behavior of an IP SLA (Service Level Agreement) when configured to monitor end-to-end delays between devices in a network?

- A. IP SLA requires both devices being monitored to be Cisco devices.
- B. IP SLA is used to monitor end-to-end delays and does not support one-way delay measurement.
- C. IP SLA can only measure round-trip delays and does not require time synchronization.
- D. IP SLA measures one-way delays and requires time synchronization mechanisms like NTP for accurate measurements.

Answer:

D. IP SLA measures one-way delays and requires time synchronization mechanisms like NTP for accurate measurements. ✅

Explanation:

IP SLA on Cisco devices can measure both **round-trip delay** and **one-way delay**. However, for accurate **one-way delay measurement**, both devices must have synchronized clocks using protocols like **NTP (Network Time Protocol)**.

Without time synchronization, only round-trip delay can be reliably measured.

Other options:

- **A** → Not strictly required; IP SLA can interact with non-Cisco devices depending on operation
- **B** → Incorrect; IP SLA DOES support one-way delay
- **C** → Incorrect; one-way delay requires time synchronization

Explanation (Simplified):

IP SLA can measure:

- Round-trip delay → No sync needed
- One-way delay → **Time sync required (NTP)**

👉 Why?

- One-way delay compares send time vs receive time
- Needs both clocks to match accurately

🧠 Quick Memory Trick:

- Round-trip = No sync needed 📩
- One-way = **Clock sync needed (NTP)** 🕒

Key Point:

One-way delay measurement = Requires NTP time synchronization 🚀

Question 14

In the context of IEEE 802.1X, which of the following statements best describes the function of the supplicant in a network access scenario?

- The supplicant is the device or software that requests access to the network and responds to the authenticator's queries.
- The supplicant is responsible for granting or denying network access to the requesting device.
- The supplicant is the network switch that enforces the security policies and controls access to the network.
- The supplicant acts as a middleman between the authenticator and the authentication server, forwarding messages back and forth.

Answer:

A. The supplicant is the device or software that requests access to the network and responds to the authenticator's queries. ✅

Explanation:

In **IEEE 802.1X**, the **supplicant** is the client device (such as a laptop or phone) or software that initiates the authentication process. It provides credentials (username, password, or certificate) when prompted by the authenticator.

Other roles:

- **Authenticator** → Switch/AP that controls access and forwards requests
- **Authentication Server** → Makes the final decision (allow/deny)

Other options:

- **B** → This is the role of the Authentication Server
- **C** → This describes the Authenticator
- **D** → Also describes the Authenticator (middleman)

Explanation (Simplified):

In IEEE 802.1X, there are three roles:

- Supplicant → The client requesting access
- Authenticator → The switch/AP that passes credentials
- Authentication Server → The decision-maker

👉 The **Supplicant**:

- Requests network access
- Sends credentials
- Responds to authentication challenges

🧠 Quick Memory Trick:

- Supplicant = **Requests access (client)** 🧑

- Authenticator = Middleman
- Authentication Server = Brain 🧠

Key Point:

Supplicant = Device that requests access and provides credentials 🚀

Question 15

In an enterprise network with multiple switches and VLANs, if a trunk link between two switches is configured with a native VLAN and the native VLAN is different on each switch, what will be the most likely outcome?

- A. Frames belonging to the native VLAN from each switch will not be tagged, potentially causing VLAN hopping or traffic loss.
- B. An error message will be logged, but the network traffic will continue to flow correctly.
- C. The trunk link will automatically negotiate the correct native VLAN using the DTP (Dynamic Trunking Protocol).
- D. STP (Spanning Tree Protocol) will block the trunk port to prevent the VLAN mismatch.

Answer:

A. Frames belonging to the native VLAN from each switch will not be tagged, potentially causing VLAN hopping or traffic loss. ✅

Explanation:

In **IEEE 802.1Q**, the **native VLAN** is the only VLAN whose frames are sent **untagged** on a trunk link.

If two switches have **different native VLANs configured**, untagged frames sent from one switch will be interpreted as belonging to a different VLAN on the other switch. This can lead to:

- Traffic being placed in the wrong VLAN
- VLAN hopping vulnerabilities
- Communication issues or traffic loss

Other options:

- **B** → A mismatch may generate warnings, but traffic will NOT flow correctly
- **C** → DTP negotiates trunking, not native VLAN IDs
- **D** → STP prevents loops, not VLAN mismatches

Explanation (Simplified):

- Native VLAN = **untagged traffic**
- If mismatch happens:

👉 Switch A sends untagged → Switch B puts it in WRONG VLAN

👉 Result:

- Security risk (VLAN hopping)
- Broken communication

🧠 Quick Memory Trick:

- Native VLAN = **No tag** 🏷️ ❌
- Mismatch = **Wrong VLAN = Problems** ⚠️

Key Point:

Native VLAN mismatch = Untagged traffic misinterpreted → Security + connectivity issues



Question 16

What is the primary purpose of the ARP (Address Resolution Protocol) in a local area network?

- A. To encrypt network traffic
- B. To route packets between different networks
- C. To translate IP addresses to MAC addresses
- D. To manage IP address assignments

Answer:

- C. To translate IP addresses to MAC addresses ✅

Explanation:

ARP (Address Resolution Protocol) is used in IPv4 networks to map a device's **IP address (Layer 3)** to its **MAC address (Layer 2)**.

When a device wants to communicate within a local network, it must know the destination's MAC address. If it only knows the IP address, it sends an ARP request to discover the corresponding MAC address.

Other options:

- **A** → Encryption is handled by protocols like IPsec/SSL
- **B** → Routing is performed by routers
- **D** → IP assignment is done by DHCP

Explanation (Simplified):

- Device knows IP → Needs MAC to send frame

👉 ARP does:

- “Who has this IP?”
- Device replies with MAC address

👉 Then communication starts

🔄 Quick Memory Trick:

- ARP = **IP** → **MAC lookup** 🔍
- “Who has this IP?” → “Here is my MAC”

Key Point:

ARP = Converts IP address to MAC address for local communication 🚀

Question 17

In a scenario where multiple routing protocols are configured on a router, which factor has the highest priority in selecting the best path to a destination network if all other metrics are equal?

- A. Delay
- B. Bandwidth
- C. Administrative distance
- D. Hop count

Answer:

C. Administrative distance 

Explanation:

When a router learns routes to the same destination from **different routing protocols**, it uses **Administrative Distance (AD)** to decide which route to trust.

- **Lower AD = More preferred route**

AD is considered **before metrics** like bandwidth, delay, or hop count when comparing routes from different protocols.

Other options:

- **A (Delay)** → Metric used by some protocols (like EIGRP)
- **B (Bandwidth)** → Metric component (EIGRP, OSPF cost indirectly)
- **D (Hop count)** → Used by RIP

👉 These metrics are only compared **within the same routing protocol**, not across different ones.

Explanation (Simplified):

When multiple protocols exist:

- Router first checks → **Administrative Distance**
- Then checks → Metrics (if same protocol)

👉 Example:

- OSPF route (AD 110) vs RIP route (AD 120)
→ Router chooses **OSPF**

🧠 Quick Memory Trick:

- AD = **Trust level of route** 🧠
- Lower AD = More trusted
- “Different protocol? → Check AD first”

Key Point:

Administrative Distance = Highest priority for route selection across different protocols 🚀

Question 18

In the context of DHCP snooping, what happens when a switch receives a DHCP OFFER from an untrusted port?

- A. The switch will mark the OFFER packet and send it to the network administrator for review.
- B. The switch will drop the DHCP OFFER packet.
- C. The switch will forward the DHCP OFFER to all VLANs.
- D. The switch will insert an option 82 field in the DHCP OFFER packet.

Answer:

- B. The switch will drop the DHCP OFFER packet. ✅

Explanation:

DHCP snooping is a security feature that prevents rogue DHCP servers from assigning IP addresses. Ports on a switch are classified as:

- **Trusted ports** → Connected to legitimate DHCP servers
- **Untrusted ports** → Connected to clients

👉 If a DHCP OFFER is received on an **untrusted port**, the switch assumes it may be from a rogue DHCP server and **drops the packet** to protect the network.

Other options:

- **A** → No such automatic review process
- **C** → DHCP traffic is not forwarded across all VLANs like that
- **D** → Option 82 is inserted on trusted ports, not related to dropping rogue offers

Explanation (Simplified):

- Trusted port = DHCP server allowed
- Untrusted port = Clients only

👉 If OFFER comes from untrusted:

-  Block it (security risk)

Quick Memory Trick:

- DHCP Snooping = “**Trust only real DHCP server**” 
- OFFER from wrong place = **DROP** 

Key Point:

Untrusted port + DHCP OFFER = Packet dropped (rogue DHCP protection) 

Question 19

In the context of network security, which of the following configurations must be implemented on a Cisco router to ensure OSPF neighbor authentication using MD5 hashing?

A.

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
interface GigabitEthernet0/0
ip ospf authentication
ip ospf authentication-key 12345
```

B.

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
interface GigabitEthernet0/0
ip ospf message-digest-key 1 md5 MySecretKey
ip ospf authentication message-digest
```

C.

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
ip ospf authentication message-digest
```

D.

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
interface GigabitEthernet0/0
ip ospf message-digest-key 1 md5 MySecretKey
ip ospf authentication-key 12345
```

Answer:

B. 

Explanation:

To configure **OSPF MD5 authentication**, you must:

1. Enable MD5 authentication on the interface
2. Configure an MD5 key with a key ID

Option **B** correctly includes both:

- `ip ospf message-digest-key 1 md5 MySecretKey` → Defines the MD5 key
- `ip ospf authentication message-digest` → Enables MD5 authentication

Other options:

- **A** → Uses plain-text authentication, not MD5
- **C** → Enables MD5 but does NOT define the key
- **D** → Mixes MD5 key with plain-text authentication (incorrect)

Explanation (Simplified):

For OSPF MD5:

👉 You need TWO things:

- Enable MD5 → authentication message-digest
- Set key → message-digest-key

👉 Missing either = Won't work ❌

🧠 Quick Memory Trick:

- MD5 = **Secure authentication** 🗝️
- 2 steps → Enable + Key

👉 “No key = No security”

Key Point:

OSPF MD5 = **message-digest + key configuration on interface** 🚀

Question 20

You have a network with subnet 192.168.1.0/25 and need to create the maximum number of equal-sized subnets with a minimum of 20 hosts each. What is the new subnet mask?

- A. /28
- B. /27
- C. /29
- D. /26

Answer:

B. /27 ✅

Explanation:

We need subnets that support **at least 20 hosts**.

👉 Use formula:

$$\text{Hosts per subnet} = 2^n - 2$$

Check options:

- /28 → 16 IPs → 14 usable ❌ (not enough)
- /27 → 32 IPs → 30 usable ✅ (fits 20 hosts)
- /26 → 64 IPs → 62 usable (works but fewer subnets)
- /29 → 8 IPs → 6 usable ❌

👉 Since question asks for **maximum number of subnets**, choose the **smallest subnet that fits 20 hosts** → /27

Explanation (Simplified):

- Need ≥ 20 hosts
- /27 gives 30 hosts → perfect

👉 Smaller subnet (/28) = not enough

👉 Bigger subnet (/26) = wastes space

👉 So best choice = /27

🧠 Quick Memory Trick:

- /27 = 32 IPs → 30 usable
- Think: “20 users → need ~30 IPs → /27”

Key Point:

Pick smallest subnet that fits hosts → /27 gives max number of subnets 🚀

Question 21

In a mixed IPv4 and IPv6 network environment, an IPv6-enabled router forwards a packet with a source IPv4 address of 192.168.1.10 to an IPv6-only destination via a tunnel. Which encapsulating method would allow successful communication in this scenario?

- A. Teredo
- B. ISATAP
- C. NAT64
- D. 6to4

Answer:

D. 6to4 

Explanation:

6to4 is a tunneling mechanism that allows IPv6 packets to be transmitted over an IPv4 network by **encapsulating IPv6 traffic inside IPv4 packets**. It is specifically designed for communication between IPv6 networks across an IPv4 infrastructure.

In this scenario, the router is using a tunnel to reach an IPv6-only destination, which matches the purpose of **6to4 tunneling**.

Other options:

- **A (Teredo)** → Used for IPv6 over IPv4 through NAT (host-based, not typical router tunnel)
- **B (ISATAP)** → Used within an internal IPv4 network (intranet), not for general internet tunneling
- **C (NAT64)** → Translation, not tunneling (IPv6 ↔ IPv4 conversion)

Explanation (Simplified):

- Need to send IPv6 traffic over IPv4 network

 Solution = Tunnel

 6to4 does:

- Wrap IPv6 inside IPv4
- Send through IPv4 network
- Unwrap at destination

Quick Memory Trick:

- 6to4 = “IPv6 over IPv4 tunnel” 
- NAT64 = Translation
- Teredo = NAT workaround
- ISATAP = Internal use

Key Point:

6to4 = Encapsulates IPv6 packets inside IPv4 for tunneling 

Question 22

Which subnet mask would you use to configure a network with 1,022 usable IP addresses?

- A. 255.255.248.0
- B. 255.255.252.0
- C. 255.255.240.0
- D. 255.255.254.0

Answer:

B. 255.255.252.0 

Explanation:

We use the formula:

 Usable hosts = $2^n - 2$

We need **1,022 usable IPs**

- $2^{10} = 1024 \rightarrow 1024 - 2 = \mathbf{1022 \text{ usable}}$

 So we need **10 host bits**

Now calculate subnet:

- Total bits = 32
- Network bits = $32 - 10 = \mathbf{/22}$

👉 Subnet mask for /22 = **255.255.252.0**

Check options:

- **255.255.252.0 (/22)** → 1022 usable ✅
- 255.255.248.0 (/21) → 2046 usable (too large)
- 255.255.240.0 (/20) → 4094 usable (too large)
- 255.255.254.0 (/23) → 510 usable (too small)

Explanation (Simplified):

- Need ~1000 hosts
- Closest power = 1024 → /22

👉 So answer = **255.255.252.0**

🧠 Quick Memory Trick:

- 1022 hosts → think **/22**
- /22 = **255.255.252.0**

Key Point:

1,022 usable IPs = **/22 subnet mask (255.255.252.0)** 🚀

Question 23

Which of the following configurations on a Cisco access port will result in the prevention of VLAN hopping attacks while allowing the necessary VLAN traffic?

- A. Enable trunk mode and set the native VLAN to the management VLAN
- B. Enable PortFast and set the native VLAN to an unused VLAN
- C. Disable DTP and enable BPDU Guard
- D. Disable DTP and set the native VLAN to the management VLAN

Answer:

C. Disable DTP and enable BPDU Guard ✅

Explanation:

VLAN hopping attacks often exploit **Dynamic Trunking Protocol (DTP)** to negotiate a trunk link and gain access to multiple VLANs.

👉 To prevent this:

- **Disable DTP** → Prevents automatic trunk negotiation
- **Enable BPDU Guard** → Protects against rogue switches sending BPDUs

This combination secures the access port while still allowing normal VLAN traffic.

Other options:

- **A** → Trunk mode increases risk of VLAN hopping
- **B** → PortFast helps STP convergence, not VLAN hopping prevention
- **D** → Native VLAN change alone does not stop DTP-based attacks

Explanation (Simplified):

VLAN hopping happens when:

- Port becomes trunk unexpectedly
- Attacker accesses multiple VLANs

👉 Prevention:

- Disable trunk negotiation (DTP)
- Protect switch from rogue devices (BPDU Guard)

🧠 Quick Memory Trick:

- VLAN hopping = **Trunk attack**
- Solution = **No trunk + secure port** 🔒

Key Point:

Disable DTP + enable BPDU Guard = Secure access port against VLAN hopping 🚀

Question 24

A network administrator needs to enhance the router security by implementing SSH (Secure Shell) for remote access. Which series of commands should be correctly used to achieve this while ensuring the highest security level?

A.

```
router(config)# hostname R1
router(config)# ip domain-name example.com
router(config)# crypto key generate rsa modulus 1024
router(config)# line vty 0 4
router(config-line)# transport input ssh
router(config-line)# login local
```

B.

```
router(config)# interface g0/0
router(config-if)# ip address 192.168.1.1 255.255.255.0
router(config-if)# no shutdown
router(config)# ip domain-name example.com
router(config)# line console 0
router(config-line)# transport input ssh
router(config-line)# login local
```

C.

```
router(config)# hostname R1
router(config)# interface g0/0
router(config-if)# ip address 192.168.1.1 255.255.255.0
router(config-if)# exit
router(config)# line vty 0 4
router(config-line)# login local
```

D.

```
router(config)# ip domain-name example.com
router(config)# crypto key generate rsa modulus 2048
router(config)# username admin password Cisc0123
router(config)# line vty 0 4
router(config-line)# transport input telnet ssh
router(config-line)# login local
```

Answer:

A. 

Explanation:

To securely configure SSH on a Cisco router, you need:

1. **Hostname + domain name** → Required for RSA key generation
2. **RSA keys** → Enable SSH encryption
3. **VTY lines configured for SSH only** → Secure remote access
4. **Local login authentication**

👉 Option **A** correctly includes all required steps and restricts access to **SSH only**, which is more secure than allowing Telnet.

Other options:

- **B** → Incorrect (SSH is not configured on console line)
- **C** → Missing RSA keys → SSH will not work
- **D** → Allows Telnet (transport input telnet ssh) → Less secure ❌

Explanation (Simplified):

To enable SSH:

- Set hostname + domain
- Generate RSA keys
- Allow only SSH on VTY
- Use local login

👉 Important:

- SSH = Encrypted 
- Telnet = Not secure ❌

Quick Memory Trick:

- SSH setup = **Name + Domain + Keys + VTY config** 

- “No keys = No SSH”

Key Point:

Secure SSH = RSA keys + SSH-only access (no Telnet) 🚀

Question 25

Consider a scenario where you need to design a small subnet for 6 devices. Given that you must use a subnet mask providing the least amount of unused IP addresses while allowing all devices to communicate within the same subnet, what subnet mask should you use?

- A. /27
- B. /28
- C. /29
- D. /30

Answer:

C. /29 ✅

Explanation:

We need a subnet that supports **at least 6 usable hosts** with minimal waste.

👉 Formula:

$$\text{Usable hosts} = 2^n - 2$$

Check options:

- /30 → 4 IPs → 2 usable ❌ (not enough)
- /29 → 8 IPs → 6 usable ✅ (perfect fit)
- /28 → 16 IPs → 14 usable (too many, wasted)
- /27 → 32 IPs → 30 usable (too many, wasted)

👉 /29 is the smallest subnet that supports exactly 6 hosts, minimizing unused IPs.

Explanation (Simplified):

- Need = 6 devices
- /29 gives = 6 usable IPs

👉 Perfect match = no waste

🧠 Quick Memory Trick:

- /30 = 2 hosts
- /29 = **6 hosts** ✅
- /28 = 14 hosts

👉 “6 hosts → think /29”

Key Point:

6 devices = **/29 subnet (8 total, 6 usable)** 🚀

Question 26

When configuring DHCP on a Cisco router, which of the following options is NOT a valid DHCP configuration command?

- A. default-router [address]
- B. network [subnet] [subnet mask]
- C. ip helper-address [address]
- D. ip dhcp pool

Answer:

C. ip helper-address [address] ✅

Explanation:

In Cisco DHCP configuration:

- **ip dhcp pool** → Creates a DHCP pool
- **network [subnet] [mask]** → Defines the subnet for the pool

- **default-router [address]** → Specifies the default gateway for clients

👉 These are all valid DHCP server configuration commands.

However:

- **ip helper-address** is **not part of DHCP server configuration**
- It is used on router interfaces to **relay DHCP requests** from clients to a remote DHCP server

Explanation (Simplified):

- DHCP server config = pool, network, gateway

👉 ip helper-address =

- Used when DHCP server is on another network
- Forwards DHCP requests

🧠 Quick Memory Trick:

- DHCP pool = Server setup
- ip helper = **Forward DHCP (relay)** 📩

Key Point:

ip helper-address = DHCP relay, NOT DHCP server configuration 🚀

Question 27

You configure 802.1X port-based authentication in an enterprise network with multiple switches. Which step is crucial to ensure devices without supplicant software can still gain limited network access for bootstrapping or management tasks?

- A. Enable Guest VLAN on the switch ports.
- B. Configure MAC Authentication Bypass (MAB) on the switch ports.
- C. Enable EAP over LAN (EAPoL) on all switch ports.
- D. Implement RADIUS server failover.

Answer:

B. Configure MAC Authentication Bypass (MAB) on the switch ports. 

Explanation:

In **IEEE 802.1X**, devices normally authenticate using a supplicant (username/certificate). However, some devices (like printers, IP phones, IoT) do not support 802.1X.

👉 **MAC Authentication Bypass (MAB)** allows these devices to authenticate using their **MAC address** instead. The switch sends the MAC address to the RADIUS server, which decides whether to allow access.

Other options:

- **A** → Guest VLAN provides limited access but does not authenticate devices
- **C** → EAPoL is part of 802.1X, but non-supplicant devices cannot use it
- **D** → RADIUS failover improves availability, not access for non-supplicant devices

Explanation (Simplified):

- 802.1X = Needs supplicant
- No supplicant → Use **MAB**

👉 MAB does:

- Uses MAC address as identity
- Allows basic/limited access
- Supports devices like printers, IP phones

Quick Memory Trick:

- No supplicant = **Use MAB** 
- MAB = “Login using MAC address”

Key Point:

MAB = Enables access for devices without 802.1X supplicant 

Question 28

Which of the following statements about OSPF (Open Shortest Path First) are correct?

- A. In OSPF, the router with the highest IP address on its loopback interface becomes the designated router (DR).
- B. OSPF sends the entire routing table to all routers at regular intervals.
- C. OSPF uses a Distance Vector algorithm for routing.
- D. OSPF allows for the separation of a network into different areas to optimize routing.

Answer:

- D. OSPF allows for the separation of a network into different areas to optimize routing. 

Explanation:

OSPF is a **link-state routing protocol** that supports hierarchical design using **areas**.

Dividing a network into areas helps:

- Reduce routing table size
- Improve performance
- Limit LSAs (Link-State Advertisements) flooding

Other options:

- **A** → Incorrect; DR election is based on **priority first**, then highest Router ID (not just loopback IP alone)
- **B** → Incorrect; OSPF sends updates only when changes occur (not periodic full table updates like RIP)
- **C** → Incorrect; OSPF uses **link-state algorithm (SPF/Dijkstra)**, not distance vector

Explanation (Simplified):

OSPF features:

- Uses **link-state (fast + efficient)**
- Sends updates only when needed
- Supports **areas for scalability**

👉 Areas help:

- Organize network
- Reduce overhead
- Improve routing efficiency

🧠 Quick Memory Trick:

- OSPF = **Link-State + Areas** 🧠
- RIP = Distance Vector
- “OSPF scales with areas”

Key Point:

OSPF = Hierarchical design using areas for better scalability 🚀

Question 29

In a large, enterprise-grade network, you have been tasked with configuring 802.1x port-based authentication. Given the complexity of the network, which of the following could potentially result in authentication failures due to security-related misconfigurations?

- A. RADIUS server certificate expired
- B. Use of dynamic VLAN assignments
- C. Switchport mode set to ‘access’
- D. Incorrect EAP method configured on the client and the server

Answer:

- A. RADIUS server certificate expired
- D. Incorrect EAP method configured on the client and the server ✅

Explanation:

In **IEEE 802.1X**, authentication relies on secure communication between the supplicant and the RADIUS server.

👉 The following can cause authentication failures:

- **Expired RADIUS certificate** → Client cannot trust server → authentication fails
- **EAP mismatch (client vs server)** → Authentication method fails (e.g., PEAP vs EAP-TLS mismatch)

Other options:

- **B** → Dynamic VLAN assignment does not cause authentication failure; it applies after successful authentication
- **C** → Access mode is correct for 802.1X on end-device ports

Explanation (Simplified):

802.1X needs:

- Valid certificate
- Matching authentication method

👉 If either fails:

- Authentication fails ❌

👉 But VLAN or port mode does NOT break authentication

🧠 Quick Memory Trick:

- 802.1X fails when:
 - ❌ Certificate invalid
 - ❌ EAP mismatch

👉 “Trust + Method must match” 🔒

Key Point:

Authentication failures = Certificate issues + EAP mismatch 🚀

Question 30

A network engineer is configuring a Cisco device for NTP authentication to ensure that time synchronization is secure. Which command should be used to authenticate the NTP messages received by the device?

- A. ntp trusted-key 1
- B. ntp server 192.168.1.1 key 1
- C. ntp authentication-key 1 md5 cisco123
- D. ntp authenticate

Answer:

D. ntp authenticate 

Explanation:

To enable **NTP authentication**, the router must be configured to actually **verify (authenticate) incoming NTP messages**.

👉 The command:

- **ntp authenticate** → Enables authentication of NTP packets

Other commands are also part of the setup but serve different roles:

- **ntp authentication-key 1 md5 cisco123** → Defines the key
- **ntp trusted-key 1** → Marks the key as trusted
- **ntp server 192.168.1.1 key 1** → Associates the key with the NTP server

👉 However, without `ntp authenticate`, authentication is **not enforced**.

Explanation (Simplified):

NTP authentication steps:

1. Create key
2. Trust key
3. Assign key to server
4. **Enable authentication** → **ntp authenticate** 

Quick Memory Trick:

- Key = Created 
- Trusted = Allowed
- Server = Uses key
- Authenticate = **Enforces security** 

Key Point:

ntp authenticate = Enables secure NTP message verification 

Question 31

In the context of IP Services, what is the primary purpose of enabling ‘ip sla responder’ on a Cisco device?

- A. To facilitate precise time synchronization between two endpoints
- B. To create VPN tunnels for secure communication
- C. To perform application-level inspection and filtering
- D. To enable enhanced ICMP echo-based performance monitoring between devices

Answer:

- D. To enable enhanced ICMP echo-based performance monitoring between devices 

Explanation:

IP SLA Responder is configured on the **target device** to respond to IP SLA probes with additional accuracy and information. It allows more precise measurement of network performance metrics such as:

- Latency (delay)
- Jitter
- Packet loss

 With `ip sla responder`, the device actively participates in the measurement process, improving accuracy compared to standard ICMP ping.

Other options:

- **A** → Time sync is handled by NTP
- **B** → VPNs use IPsec, not IP SLA
- **C** → Application inspection is done by firewalls

Explanation (Simplified):

- IP SLA = Measure network performance
- Responder = **Helps measure accurately**

👉 Without responder → basic ping

👉 With responder → detailed performance data

🧠 Quick Memory Trick:

- SLA = **Measure network health** 📊
- Responder = **Better accuracy** 🎯

Key Point:

ip sla responder = Improves accuracy of performance monitoring (delay, jitter, loss) 🚀

Question 32

In a complex OSPF network, an engineer notices that certain routers in Area 0 (Backbone Area) are not receiving the OSPF Database Description (DBD) packets properly. Which of the following could be the reason for this issue?

- A. The DBD packet always requires MD5 authentication which was not configured.
- B. The MTU size is mismatched on neighboring router interfaces.
- C. The OSPF router IDs are duplicated on neighboring routers.
- D. The 'no ip ospf shut' command was not configured correctly.

Answer:

- B. The MTU size is mismatched on neighboring router interfaces. ✅

Explanation:

In OSPF, **Database Description (DBD) packets** are exchanged during adjacency formation (EXSTART/EXCHANGE states).

👉 If there is an **MTU mismatch** between two routers:

- They cannot properly exchange DBD packets
- Adjacency may get stuck in **EXSTART or EXCHANGE state**
- Routers fail to synchronize their LSDB (Link-State Database)

Other options:

- **A** → MD5 authentication is optional, not required for DBD packets
- **C** → Duplicate Router IDs cause issues, but not specifically DBD exchange failure
- **D** → Invalid/incorrect command (not relevant)

Explanation (Simplified):

OSPF neighbors must match:

- MTU
- Area
- Authentication (if used)

👉 If MTU mismatch:

- DBD packets fail
- Adjacency stuck ❌

🧠 Quick Memory Trick:

- OSPF stuck in EXSTART/EXCHANGE → **Check MTU first** 🔍
- “No MTU match = No adjacency”

Key Point:

MTU mismatch = DBD exchange failure → OSPF adjacency issue 🚀

Question 33

Which feature of Cisco's Identity Services Engine (ISE) allows for dynamically assigning user roles, permissions, and network access based on contextual data, such as user identity, location, and security posture?

- A. TrustSec
- B. Posture Assessment
- C. Network Admission Control (NAC)
- D. Context-Aware Access

Answer:

D. Context-Aware Access 

Explanation:

Cisco ISE Context-Aware Access uses multiple factors (context) such as:

- User identity
- Device type
- Location
- Security posture

👉 Based on this information, it dynamically assigns:

- Access permissions
- Security policies
- Network roles

This enables **fine-grained, intelligent access control** instead of static rules.

Other options:

- **A (TrustSec)** → Focuses on segmentation using Security Group Tags (SGTs)
- **B (Posture Assessment)** → Checks device compliance (antivirus, patches, etc.)
- **C (NAC)** → General concept of controlling network access, not context-driven decisions

Explanation (Simplified):

Context-aware access =

👉 “Who + Where + Device + Security status”

→ Decide access level dynamically

Example:

- Employee laptop in office → Full access
- Unknown device → Limited access
- Infected device → Blocked

Quick Memory Trick:

- Context-aware = **Smart access decisions** 🧠
- Based on **identity + situation**

Key Point:

Context-Aware Access = Dynamic, intelligent access control using multiple factors 🚀

Question 34

In the context of implementing DHCP, which command would you use on a Cisco router to exclude a range of addresses from being assigned to clients, and why is this step necessary in a well-designed network?

- A. ip dhcp pool MyPool
- B. ip dhcp server pool 192.168.1.0 255.255.255.0
- C. ip dhcp excluded-address 192.168.1.1 192.168.1.10
- D. ip dhcp excluded-range 192.168.1.1 192.168.1.10

Answer:

C. ip dhcp excluded-address 192.168.1.1 192.168.1.10 ✅

Explanation:

The command **ip dhcp excluded-address** is used to prevent specific IP addresses (or a range) from being assigned by the DHCP server.

👉 This is necessary because some IP addresses are typically reserved for:

- Default gateway
- Servers (DNS, web, file servers)
- Network devices (routers, switches, printers)

If these addresses are not excluded, DHCP could assign them to clients, causing **IP conflicts** and network issues.

Other options:

- **A** → Creates a DHCP pool
- **B** → Invalid command
- **D** → Not a valid Cisco command

Explanation (Simplified):

- DHCP gives IPs automatically

👉 But some IPs must be reserved

👉 So we **exclude them**

Example:

- Router = 192.168.1.1
- Server = 192.168.1.2

👉 Exclude these → avoid conflicts

Quick Memory Trick:

- Excluded = “**Do NOT give these IPs**” 🚫
- Prevents IP conflicts

Key Point:

ip dhcp excluded-address = Reserve IPs and prevent DHCP conflicts 🚀

Question 35

In a network utilizing Enhanced Interior Gateway Routing Protocol (EIGRP), which attribute does EIGRP use by default to calculate the best path to a remote network?

- A. Hop Count
- B. Cost
- C. MTU
- D. Bandwidth and Delay

Answer:

D. Bandwidth and Delay ✅

Explanation:

EIGRP uses a composite metric by default, which is primarily based on:

- **Bandwidth** (minimum along the path)
- **Delay** (cumulative along the path)

These two values are combined to calculate the best path.

Other metrics like reliability and load exist but are **not used by default**.

Other options:

- **A (Hop Count)** → Used by RIP
- **B (Cost)** → Used by OSPF
- **C (MTU)** → Not used in EIGRP metric calculation

Explanation (Simplified):

EIGRP chooses best path based on:

- Fastest link (Bandwidth)

- Least delay (Delay)

👉 More bandwidth + less delay = better route

Quick Memory Trick:

- EIGRP = “**Best path = Speed + Delay**” ⚡
- RIP = Hop count
- OSPF = Cost

Key Point:

EIGRP metric = **Bandwidth + Delay (default)** 🚀

Question 36

In a network security context, what is the primary significance of implementing 802.1X Port-Based Network Access Control (PNAC) on switch ports?

- A. It ensures that all users must change their passwords every 30 days.
- B. It encrypts all traffic on the network to ensure confidentiality.
- C. It monitors and logs inbound and outbound traffic for security purposes.
- D. It helps prevent unauthorized devices from connecting to the network, by authenticating each device before it is granted access.

Answer:

D. It helps prevent unauthorized devices from connecting to the network, by authenticating each device before it is granted access. ✅

Explanation:

IEEE 802.1X (PNAC) is used to **control access at the port level** by requiring devices to authenticate before gaining network access.

👉 It ensures that:

- Only authorized users/devices can connect
- Unauthorized devices are blocked

- Network access is granted based on authentication results

Other options:

- **A** → Password policies are handled by identity management systems
- **B** → Encryption is handled by protocols like IPsec/SSL
- **C** → Traffic monitoring is done by IDS/IPS or firewalls

Explanation (Simplified):

802.1X works like a gate:

- Device connects → must authenticate
- If valid → access granted
- If not → access denied ❌

🧠 Quick Memory Trick:

- 802.1X = “No login = No network” 🚫
- Authentication first, access later

Key Point:

802.1X PNAC = Prevents unauthorized access by enforcing authentication 🚀

Question 37

In an IPv6 network, which of the following IPv6 addresses is used as the all-nodes multicast address that can be sent to all nodes on the local network segment?

- A. FF02::1
- B. FF02::A
- C. FF02::5
- D. FF02::2

Answer:

- A. FF02::1 ✅

Explanation:

In IPv6, **multicast addresses** are used to send packets to multiple devices.

👉 **FF02::1** is the **all-nodes multicast address**, meaning:

- It reaches **all IPv6-enabled devices** on the local link
- It is similar to a broadcast in IPv4, but more efficient

Other options:

- **FF02::2** → All routers on the local link
- **FF02::5** → OSPF routers
- **FF02::A** → EIGRP routers

Explanation (Simplified):

- FF02::1 = **All devices** 👥
- FF02::2 = Routers only

👉 “Send to everyone → use FF02::1”

🧠 Quick Memory Trick:

- **::1 = Everyone** 🌐
- **::2 = Routers**

👉 “1 = All nodes”

Key Point:

FF02::1 = All-nodes multicast address in IPv6 🚀

Question 38

Which of the following descriptions correctly explains the difference between NAT, PAT, and Proxy, especially in the context of IP address conservation and network security?

A. NAT conserves addresses by assigning a new IP address for each new internal connection, PAT allows a single public address to be translated dynamically, and Proxy servers provide IP conservation by creating temporary IP address pools.

B. NAT uses dynamic IP translation to optimize addressing, PAT translates multiple public IP addresses to a single private internal IP, and Proxy servers use static address translation to hide internal IPs.

C. NAT translates static IP addresses from private to public, PAT dynamically assigns IP addresses within a subnet, and Proxy servers are solely responsible for the routing and switching of traffic within a private network.

D. NAT translates private internal IP addresses to a single public external IP address, PAT uses unique combinations of IP addresses and port numbers, while Proxy translates between private and public IP addresses and hides internal network structure from the external network.

Answer:

D. 

Explanation:

- **NAT (Network Address Translation)** → Translates private IP addresses to public IP addresses (can be one-to-one or many-to-one).
- **PAT (Port Address Translation)** → A form of NAT that allows **multiple devices to share one public IP** using different port numbers.
- **Proxy Server** → Acts as an intermediary between internal clients and external servers, hiding internal IPs and adding a layer of security.

👉 Option **D** correctly describes all three concepts:

- NAT → IP translation
- PAT → IP + port (many-to-one)
- Proxy → Hides internal network + adds security

Other options are incorrect because they misunderstand how NAT/PAT work or confuse their roles.

Explanation (Simplified):

- NAT = Change IP
- PAT = Share 1 IP using ports
- Proxy = Middleman (security + hiding)

👉 Example:

- 100 users → 1 public IP (PAT)
- Proxy → users don't directly talk to internet

🧠 Quick Memory Trick:

- NAT = **IP change** 🔄
- PAT = **Many → 1 (ports)**

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34
- Proxy = **Middleman + security** 🛡️

Key Point:

PAT = IP + Port sharing, Proxy = hides internal network, NAT = IP translation 🚀

Question 39

In a complex enterprise network, you are asked to design a redundant setup for DHCP services to ensure high availability. Which of the following options best describes a recommended setup for DHCP failover using Cisco routers?

- A. Configure DHCP servers in a split-scope configuration with 50:50 allocation on two different routers.
- B. Configure two DHCP servers with identical scopes and set up HSRP for failover.
- C. Implement DHCP failover in load balance mode, splitting the load but synchronizing lease information between two servers.
- D. Use a DHCP relay agent to forward requests to two different DHCP servers and allow the clients to use whichever server responds first.

Answer:

C. Implement DHCP failover in load balance mode, splitting the load but synchronizing lease information between two servers. 

Explanation:

The **best practice for DHCP high availability** is to use **DHCP failover with load balancing**, where:

- Two DHCP servers share the workload
- Lease information is **synchronized between both servers**
- If one server fails, the other continues seamlessly

 This ensures:

- No IP conflicts
- Continuous service
- Efficient load distribution

Other options:

- **A** → Split-scope works but is outdated and not fully synchronized
- **B** → HSRP is for gateway redundancy, not DHCP synchronization
- **D** → Can cause IP conflicts and inconsistent leases (not reliable)

Explanation (Simplified):

Best setup =

- 2 DHCP servers
- Share load
- Sync data

 If one fails → other continues smoothly

Quick Memory Trick:

- DHCP HA = **Sync + Share load** 

- No sync = Risk of IP conflicts ❌

Key Point:

DHCP failover (load balance + sync) = Best high availability solution 🚀

Question 40

What is the primary function of Network-Based Access Control (NBAC) in a Cisco network environment?

- A. It provides Layer 2 security by establishing VLANs for different types of access levels.
- B. It dynamically assigns IP addresses to hosts within the network.
- C. It controls access at the network layer by using rules that allow or deny traffic based on IP addresses, protocols, and ports.
- D. It monitors network traffic for suspicious activities and generates alerts for potential threats.

Answer:

C. It controls access at the network layer by using rules that allow or deny traffic based on IP addresses, protocols, and ports. ✅

Explanation:

Network-Based Access Control (NBAC) is used to enforce security policies by controlling which traffic is allowed or denied in a network. It operates mainly at **Layer 3/4** and uses criteria such as:

- IP addresses
- Protocols (TCP/UDP/ICMP)
- Port numbers

👉 This is typically implemented using **Access Control Lists (ACLs)** on routers and firewalls.

Other options:

- **A** → VLANs are Layer 2 segmentation, not NBAC

- **B** → DHCP handles IP assignment
- **D** → IDS/IPS systems monitor and alert, not control access directly

Explanation (Simplified):

NBAC =

👉 “Who can talk to whom?”

- Allow traffic ✅
- Deny traffic ❌

Based on:

- IP
- Port
- Protocol

🧠 Quick Memory Trick:

- NBAC = **Network traffic control (ACLs)** 🚦
- Allow / Deny rules

Key Point:

NBAC = Controls traffic using IP, port, and protocol rules 🚀

Question 41

In a Cisco switch with multiple VLANs configured, which dynamic trunking protocol (DTP) mode essentially prevents the port from generating DTP frames but allows it to respond to DTP frames from neighboring devices?

- A. Dynamic Desirable
- B. Trunk
- C. Access
- D. Dynamic Auto

Answer:

D. Dynamic Auto 

Explanation:

Dynamic Auto mode in DTP means the port:

- **Does NOT actively send DTP frames**
- **Listens and responds** to DTP frames from the neighboring device
- Will become a trunk only if the other side initiates it (e.g., Dynamic Desirable or Trunk)

Other options:

- **A (Dynamic Desirable)** → Actively sends DTP frames to form a trunk
- **B (Trunk)** → Forces trunking, does not rely on DTP negotiation
- **C (Access)** → Disables trunking and DTP negotiation entirely

Explanation (Simplified):

DTP modes:

- Dynamic Desirable → **Actively tries trunk**
- Dynamic Auto → **Waits and responds** 

 Auto = Passive mode

Quick Memory Trick:

- Auto = **Wait mode** 
- Desirable = **Try to trunk**

 “Auto listens, doesn’t talk first”

Key Point:

Dynamic Auto = Does not initiate DTP, only responds 

Question 42

In a segmented network using VLANs, what additional security risk must be mitigated to ensure inter-VLAN communication does not create vulnerabilities?

- A. ARP spoofing
- B. VLAN hopping
- C. MAC address flooding
- D. DHCP starvation

Answer:

- B. VLAN hopping 

Explanation:

VLAN hopping is a security attack where an attacker gains access to traffic from other VLANs, bypassing VLAN segmentation. This directly threatens **inter-VLAN communication security**.

👉 It can occur through:

- Switch spoofing (tricking a port into becoming a trunk)
- Double tagging (manipulating VLAN tags)

Other options:

- **A (ARP spoofing)** → Affects IP-to-MAC mapping, not VLAN boundaries
- **C (MAC flooding)** → Overloads CAM table, not VLAN crossing
- **D (DHCP starvation)** → Exhausts IP pool, not VLAN segmentation issue

Explanation (Simplified):

VLANs = Separate networks

👉 But attacker tries to:

- Jump between VLANs = VLAN hopping 

👉 Must prevent to keep VLANs secure

Quick Memory Trick:

- VLAN hopping = “Jump VLANs” 
- Breaks network separation

Key Point:

VLAN hopping = Main security risk in VLAN-based networks 

Question 43

Which Cisco feature can be used to ensure that only authenticated devices are allowed to connect to the network through a switch port?

- A. Port Security
- B. Access Control Lists (ACLs)
- C. 802.1X
- D. DHCP Snooping

Answer:

C. 802.1X 

Explanation:

IEEE 802.1X provides **port-based authentication**, ensuring that devices must authenticate before gaining access to the network.

 It works by:

- Requiring credentials (username/password/certificate)
- Verifying via RADIUS server
- Allowing or denying access

Other options:

- **A (Port Security)** → Limits MAC addresses, not true authentication
- **B (ACLs)** → Filters traffic, not device authentication
- **D (DHCP Snooping)** → Prevents rogue DHCP servers, not authentication

Explanation (Simplified):

- Device connects → must login
- If authenticated → access granted
- If not → blocked ❌

🧠 Quick Memory Trick:

- 802.1X = “Login before network access” 🔒
- No login = No access

Key Point:

802.1X = Ensures only authenticated devices can connect 🚀

Question 44

When configuring a switchport for Voice VLAN, which combination of commands ensures proper segregation of voice and data traffic while optimizing for IP phone connectivity?

- A. switchport mode trunk
- B. vlan VOICE_VLAN_ID; switchport trunk encapsulation dot1q
- C. switchport voice vlan; switchport priority extend cos 5
- D. switchport mode access; switchport voice vlan

Answer:

- D. switchport mode access; switchport voice vlan ✅

Explanation:

For IP phones, a switchport must support **both data and voice traffic**:

- **Access VLAN** → For PC/data traffic
- **Voice VLAN** → For IP phone traffic (tagged)

👉 Correct configuration:

- switchport mode access → Sets port for end devices

- `switchport voice vlan X` → Assigns voice VLAN for IP phone

This allows:

- Data traffic = untagged (access VLAN)
- Voice traffic = tagged (voice VLAN)

Other options:

- **A** → Trunk mode not recommended for end devices
- **B** → Incorrect/incomplete config
- **C** → QoS setting only, not full VLAN setup

Explanation (Simplified):

IP phone port = 2 VLANs:

- Data (PC)
- Voice (phone)

👉 Commands needed:

- Access mode
- Voice VLAN

🧠 Quick Memory Trick:

- Phone port = **Access + Voice VLAN** 📞
- Data = untagged
- Voice = tagged

Key Point:

Access mode + voice VLAN = Proper IP phone + data separation 🚀

Question 45

In a complex enterprise network, you notice that R1, which is running OSPF, is unable to establish a neighbor relationship with R2, even though they are on the same subnet and

their interfaces are operational. After verifying the IP addressing and subnet mask configurations, what is a highly probable cause of this problem?

- A. Authentication mismatch in OSPF configurations
- B. Mismatched MTU size on the interfaces
- C. Incorrect OSPF network type configured
- D. Different OSPF Hello and Dead intervals

Answer:

- D. Different OSPF Hello and Dead intervals 

Explanation:

For OSPF neighbors to form adjacency, several parameters must match, including:

- Area ID
- Authentication (if configured)
- Hello and Dead intervals
- MTU (for full adjacency)

👉 If **Hello and Dead timers do not match**, routers will **not even become neighbors**, preventing adjacency formation entirely.

Other options:

- **A** → Also a possible issue, but typically explicitly configured
- **B** → Causes adjacency to get stuck (EXSTART/EXCHANGE), not prevent initial neighbor formation
- **C** → Can cause issues, but less common than timer mismatch in this scenario

Explanation (Simplified):

OSPF neighbors must agree on:

- Timing (Hello/Dead)

👉 If timers mismatch:

- Routers ignore each other 

- No neighbor relationship

Quick Memory Trick:

- OSPF = **Must match timers** ⌚
- No match = No neighbor

Key Point:

Hello/Dead mismatch = OSPF neighbors will not form 🚀

Question 46

Which of the following is a characteristic of 802.1X in a wired LAN?

- A. It allows for dynamic VLAN assignment based on user authentication.
- B. It requires a pre-shared key for connection establishment.
- C. It supports only EAP-MD5 for authentication.
- D. It applies only to wireless networks.

Answer:

- A. It allows for dynamic VLAN assignment based on user authentication. ✅

Explanation:

IEEE 802.1X supports **dynamic VLAN assignment**, where users are placed into different VLANs based on their authentication credentials and policies from a RADIUS server.

👉 This enables:

- Role-based access
- Network segmentation
- Improved security

Other options:

- **B** → Pre-shared keys are used in WPA/WPA2, not 802.1X
- **C** → 802.1X supports multiple EAP methods (EAP-TLS, PEAP, etc.), not just MD5

- **D** → 802.1X is used in both wired and wireless networks

Explanation (Simplified):

802.1X can:

- Authenticate user/device
- Assign VLAN dynamically

👉 Example:

- Employee → VLAN 10
- Guest → VLAN 20

🧠 Quick Memory Trick:

- 802.1X = **Login → Assign VLAN** 🗝️
- Smart network access control

Key Point:

802.1X enables dynamic VLAN assignment based on authentication 🚀

Question 47

You have just configured OSPF on a new network segment. Upon checking the OSPF neighbor table, you notice that one of the routers included in the segment does not transition from INIT state to 2-WAY state. What is the most likely cause?

- A. Mismatched subnet masks on the connected interfaces
- B. Lack of OSPF authentication configuration
- C. Misconfigured OSPF network types on two connected routers
- D. Incorrect interface MTU settings causing fragmentation issues

Answer:

- C. Misconfigured OSPF network types on two connected routers ✅

Explanation:

In OSPF, the **INIT → 2-WAY transition** happens when routers **see their own Router ID in the neighbor's Hello packet**.

👉 If routers are stuck in **INIT state**, it usually means:

- They are receiving Hellos
- But not forming a proper bidirectional relationship

One common cause is **mismatched OSPF network types** (e.g., one side broadcast, other point-to-point), which affects how Hellos are processed and neighbor relationships are formed.

Other options:

- **A** → Would prevent communication entirely (no neighbor seen)
- **B** → Authentication mismatch would block adjacency completely
- **D** → MTU issues affect EXSTART/EXCHANGE, not INIT

Explanation (Simplified):

OSPF states:

- INIT = “I see you”
- 2-WAY = “You see me too”

👉 If stuck in INIT:

- One side doesn't recognize the other properly

👉 Likely issue = Network type mismatch

🧠 Quick Memory Trick:

- INIT = One-way communication ❌
- 2-WAY = Two-way communication ✅

👉 “Not 2-way? → Check network type”

Key Point:

INIT stuck = Network type mismatch or unidirectional recognition 🚀

Question 48

You have implemented a DHCP server on a Cisco router, and clients correctly obtain IP addresses. However, clients fail to register their DNS names. What could be the cause?

- A. The DHCP configuration is missing the 'dns-server' statement.
- B. The 'ip dhcp use vrf' command has not been configured.
- C. The DHCP configuration is missing the 'option 15' command.
- D. The DHCP server does not support the feature for dynamic DNS update.

Answer:

- C. The DHCP configuration is missing the 'option 15' command. ✅

Explanation:

Option 15 (domain-name) in DHCP provides the **DNS domain name** to clients. This is essential for proper **DNS registration and name resolution**.

👉 Without Option 15:

- Clients may get IP addresses
- But cannot properly register or resolve their **fully qualified domain names (FQDNs)**

Other options:

- **A** → DNS server is needed for resolution, but not specifically for domain registration naming
- **B** → VRF is unrelated
- **D** → Cisco DHCP does support DNS-related configurations; this is not the issue

Explanation (Simplified):

- DHCP gives IP → Works ✅
- But DNS name needs domain info

👉 Missing domain = No proper DNS registration

Quick Memory Trick:

- Option 15 = **Domain name for DNS** 
- No domain = No proper DNS name

Key Point:

Option 15 (domain-name) = Required for proper DNS name registration 

Question 49

Which security feature on Cisco devices enables the enforcement of policies that can restrict and control the network traffic based on Layer 7 application information?

- A. Access Control List (ACL)
- B. Application Visibility and Control (AVC)
- C. Cisco Identity Services Engine (ISE)
- D. Zone-Based Policy Firewall (ZPF)

Answer:

B. Application Visibility and Control (AVC) 

Explanation:

Application Visibility and Control (AVC) allows Cisco devices to **identify, monitor, and control traffic based on Layer 7 (application-level) information**.

👉 It provides:

- Deep Packet Inspection (DPI)
- Application identification (e.g., YouTube, Skype, Netflix)
- Granular policy enforcement based on application type

Other options:

- **A (ACL)** → Works at Layer 3/4 (IP, port), not Layer 7

- **C (ISE)** → Focuses on identity-based access control
- **D (ZPF)** → Firewall policies, but not deep application-level visibility like AVC

Explanation (Simplified):

- AVC = **See + control applications**

👉 Example:

- Block YouTube
- Limit Netflix bandwidth
- Allow business apps only

🧠 Quick Memory Trick:

- AVC = **Application-level control (Layer 7)** 🧩
- ACL = IP/Port only
- AVC = Smart traffic control

Key Point:

AVC = Controls traffic based on application (Layer 7) 🚀

Question 50

Which of the following ACL types would provide the most granular control over the network traffic by considering both source and destination IP addresses, ports, and protocols, and why would it be used in a high-security environment?

- A. Reflexive ACLs
- B. Extended ACLs
- C. Standard ACLs
- D. Time-based ACLs

Answer:

- B. Extended ACLs ✅

Explanation:

Extended ACLs provide the most granular control because they can filter traffic based on:

- Source IP address
- Destination IP address
- Protocol (TCP, UDP, ICMP)
- Port numbers (e.g., HTTP 80, HTTPS 443)

👉 This makes them ideal for **high-security environments**, where precise control over traffic is required.

Other options:

- **A (Reflexive ACLs)** → Dynamic, stateful-like behavior but not as granular as extended ACLs
- **C (Standard ACLs)** → Only filter by source IP (very limited)
- **D (Time-based ACLs)** → Control traffic based on time, not granularity

Explanation (Simplified):

- Standard ACL = Source only ❌
- Extended ACL = **Source + Destination + Port + Protocol** ✅

👉 More control = More security

🧠 Quick Memory Trick:

- Extended ACL = **“Everything control”** 🗝️
- Standard ACL = Basic filter

👉 “Need high security → use Extended ACL”

Key Point:

Extended ACL = Most granular traffic control (IP + port + protocol) 🚀